Bao Nguyen	ECE332 Homework 1	04/09/2026 , Due on 21st

(15pts) 1.) 1. The loop shown above is in the x-y plane and B = ˆzB_0 sin ωt. What is the direction of I (ˆϕ or − ˆϕ) at:

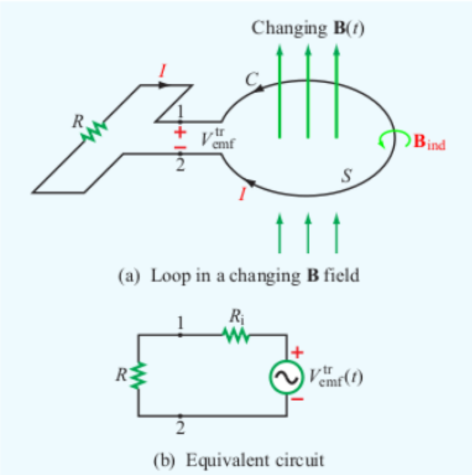

Figure 6-2 (a) Stationary circular loop in a changing magnetic field B(t), and (b) its equivalent circuit.

(a) t = 0

(b) $\omega t = \pi/4$

(c) $\omega t = \pi/2$

Notes

B — Magnetic Field

The actual magnetic field itself
Tells you the strength and direction of the field
Units: Tesla (T)
Example: $B = \hat{z}\, B_0 \sin(\omega t)$

Flux (Φ) — Magnetic Flux

How much of the magnetic field is passing through the loop
Depends on both the field strength and the area of the loop
Think of it like: how much water is flowing through a net
Units: Weber (Wb)
Formula: $\Phi = \iint B \cdot ds$

EMF (V_emf) — Electromotive Force

The voltage created by a changing flux
It is what drives the current to flow in the loop
Only exists when flux is changing
Units: Volts (V)
Formula: $V_{emf} = -d\Phi/dt$

The chain that connects them all:

B → creates Φ → changing Φ creates EMF → EMF drives current I

One key thing to remember:

B changing → Φ changing → EMF exists → current flows
B not changing → Φ not changing → EMF = 0 → no current

(14pts) 2.)

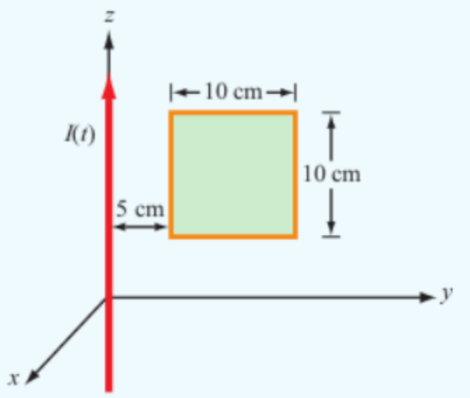

Figure P6.6 Loop coplanar with long wire (Problem 6.6).

The square loop shown above is coplanar with a long, straight wire carrying a current

$$I(t) = 5\cos\left(\left(2\pi \times 10^4 \text{ rad/s}\right) t\right) \text{ A.}$$

(a) Determine the emf induced across a small gap created in the loop. (8 points)

(b) Determine the direction and magnitude of the current that would flow through a 4Ω resistor connected across the gap. The loop has an internal resistance of 1Ω. (6 points)

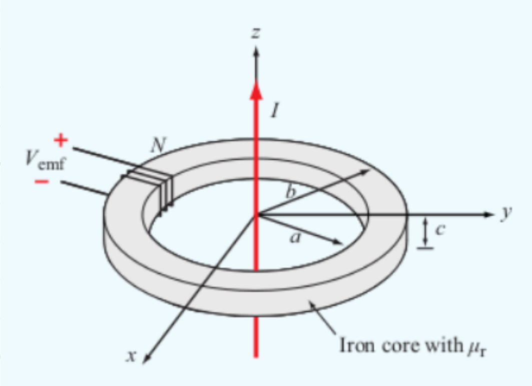

Figure P6.8 Problem 6.8.

(15pt) 3.) The transformer shown above consists of a long wire coincident with the z axis carrying a current $I = I_0 \cos \omega t$, coupling magnetic energy to the toroidal coil situated in the $x - y$ plane and centered on the origin. The toroidal core uses iron material with relative permeability μ_r, around which 100 turns of a tightly wound coil serves to induce a voltage V_{emf}, as shown in the figure.

a) Develop an expression for V_{emf}. (10 points)

(b) Calculate V_{emf} for $f = 60$ Hz, $\mu_r = 4000$, $a = 5$ cm, $b = 6$ cm, $c = 2$ cm, and $I_0 = 50$ A. (5 points)

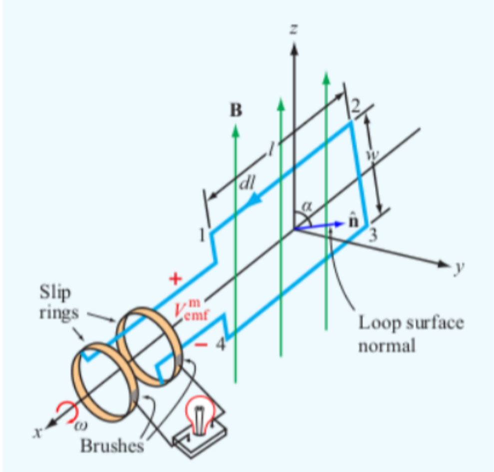

Figure 6-12 A loop rotating in a magnetic field induces an emf.

(6pts)

4. The electromagnetic generator shown above is connected to an electric bulb with a resistance of 150Ω. If the loop area is 0.1 m^2 and it rotates at 3,600 revolutions per minute in a uniform magnetic flux density B_0 = 0.4 T, determine the amplitude of the current generated in the light bulb.

(6pts) 5. The plates of a parallel-plate capacitor have areas of 10 cm^2 each and are separated by 2 cm. The capacitor is filled with a dielectric material with $\epsilon = 4\epsilon_0$, and the voltage across it is given by $V(t) = 30 \cos((2\pi \times 10^6 \text{ rad/s})\, t)$ V. Find the displacement current.

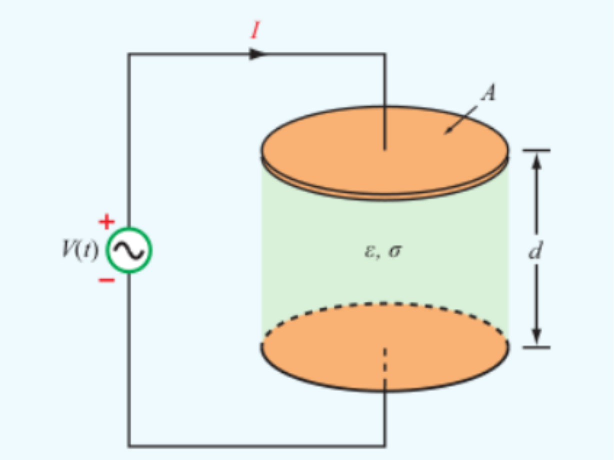

Figure P6.16 Parallel-plate capacitor containing a lossy dielectric material (Problem 6.16).

(24pts) 6. The parallel-plate capacitor shown above is filled with a lossy dielectric material of relative permittivity ϵ_r and conductivity σ. The separation between the plates is d and each plate is of area A. The capacitor is connected to a time-varying voltage source V (t).

(a) Obtain an expression for I_c, the conduction current flowing between the plates inside the capacitor, in terms of the given quantities. (6 points)

(b) Obtain an expression for I_d, the displacement current flowing inside the capacitor. (6 points)

(c) Based on your expressions for parts (a) and (b), give an equivalent-circuit representation for the capacitor. (6 points)

(d) Evaluate the values of the circuit elements for $A = 4\ cm^2$, $d = 0.5$ cm, $\epsilon_r = 4$, $\sigma = 2.5$ S/m, and $V(t) = 10 \cos((3\pi \times 10^3 \text{ rad/s})t)$. (6 points)

(6 points) 7.) At $t = 0$, charge density ρ_{v0} was introduced into the interior of a material with a relative permittivity $\epsilon_r = 9$, If at $t = 1\mu s$ the charge density has dissipated down to $10^{-3} \rho_{v0}$, what is the conductivity of the material?

(6pts)

8.) If we were to characterize how good a material is an an insulator by its resistance to dissipating charge, which of the following two materials is the better insulator?

Dry Soil:	$\epsilon_r = 2.5,$	$\sigma = 10^{-4}$ S/m
Fresh Water:	$\epsilon_r = 80,$	$\sigma = 10^{-3}$ S/m

(6pt)

9.) The electric field of an electromagnetic wave propagating in air is given by

$$\mathbf{E}(z,t) = \Big(4\cos\big((6\times10^8\ \mathrm{rad/s})\,t - (2\ \mathrm{m}^{-1})z\big)\,\hat{\mathbf{x}} \\ + 3\sin\big((6\times10^8\ \mathrm{rad/s})\,t - (2\ \mathrm{m}^{-1})z\big)\,\hat{\mathbf{y}}\Big)\ \mathrm{V/m}.$$

Find the associated magnetic field H(z, t).

(8pts)

10. Given an electric field

$$\mathbf{E} = E_0 \sin(ay) \cos(\omega t - kz)\hat{\mathbf{x}},$$

where E_0, a, ω, and k are constants, find H. (8 points)